

Tools of the Trade: Linux and SQL

Activity: Get Help in the Command Line

Introduction

In this activity, I learned how to find information about Linux commands directly from the command line.

I practiced using `whatis`, `man`, and `apropos` to understand what commands do, find command options, and search for commands when I did not remember their names.

These commands are useful because it is not always necessary to memorize every Linux command or option. The command line itself provides ways to find the information needed.

Activity Scenario

The lab focused on finding information about different Linux commands.

I first used `whatis` to get short descriptions of commands and `man` to get more detailed information.

I also used `apropos` to search for commands based on keywords.

During the activity, I worked with commands such as `cat`, `useradd`, `rm`, `rmdir`, and `groupadd`.

Lab Environment

- **Course:** Google Cybersecurity Certificate
 - **Course:** Tools of the Trade: Linux and SQL
 - **Activity:** Get Help in the Command Line
 - **Operating System:** Linux
 - **Shell:** Bash
 - **User:** `analyst`
-

Task 1: Learn More About Commands

The first task was to find a short description of the `cat` command.

I used:

```
whatis cat
```

The description showed that `cat` is used to concatenate files and display their contents.

The first two words of the description were:

concatenate files

I then used the `man` command to get more detailed information:

```
man cat
```

The manual page showed the description of the command and the different options available.

One of the questions was about numbering the output lines of `cat`.

The option for numbering all output lines is:

```
-n, --number
```

After checking the manual page, I pressed `q` to exit.

Using `apropos`

The next task was to find a command that displays the first part of a file.

I used:

```
apropos -a first part file
```

The search results helped identify the correct command.

The command that displays the first part of a file is:

```
head
```

This was useful because I did not need to remember the command name beforehand. I could search for it using keywords.

Task 2: Explore the `useradd` Command

The next task was to find an option for the `useradd` command that can set an expiration date for a temporary user account.

I opened the manual page:

```
man useradd
```

After checking the available options, I found that the option used to set an expiration date is:

```
-e
```

This showed me how the `man` command can be used when I need to understand the options available for a command.

After reviewing the information, I pressed `q` to exit the manual page.

Task 3: Explore `rm` and `rmdir`

The next task was to understand the difference between `rm` and `rmdir`.

I used:

```
whatis rm
```

and:

```
whatis rmdir
```

These short descriptions helped me compare the two commands.

The important difference is that:

```
rmdir
```

is used to remove **empty directories**.

The `rm` command is generally used to remove files and can also be used with options to remove directories.

This helped me understand when each command should be used.

Task 4: Find a Command Using Keywords

The final task was to find the command used to create a new group.

I used `apropos` with keywords:

```
apropos -a create new group
```

The search results helped me identify the correct command:

```
groupadd
```

The `groupadd` command is used to create a new group in Linux.

This was another example of how `apropos` can help when I know what I want to do but cannot remember the exact command.

Commands Practiced

Command	What I used it for
<code>whatis</code>	Get a short description of a command
<code>man</code>	Read detailed information about a command
<code>apropos</code>	Search for commands using keywords
<code>cat</code>	Display file contents
<code>useradd</code>	Add a Linux user
<code>rm</code>	Remove files
<code>rmdir</code>	Remove empty directories
<code>groupadd</code>	Create a group
<code>head</code>	Display the beginning of a file

What I Learned

This activity taught me how to find help directly from the Linux command line.

I learned that `whatis` can provide a quick description, while `man` provides more detailed information about a command and its options.

I also learned that `apropos` can be useful when I know what I want to do but do not remember the exact command.

This means I do not have to memorize every Linux command before being able to work with the terminal.

Cybersecurity Connection

Being able to find information quickly is important when working in cybersecurity.

Security analysts work with many Linux commands and tools, and it is not realistic to remember every option for every command.

The `man`, `whatis`, and `apropos` commands can help when troubleshooting a system, investigating files, managing users, or working with security tools.

Knowing how to find reliable command information directly from the system is a useful Linux skill.

Demonstration

The screenshots below show my practical work during the lab, including using `whatis`, viewing `man` pages, searching with `apropos`, and finding the required command options.

```
analyst@db9351ee59fc:~$ whatis cat
cat (1)          - concatenate files and print on the standard output
analyst@db9351ee59fc:~$ man cat
CAT(1)           User Commands                                CAT(1)

NAME
    cat - concatenate files and print on the standard output

SYNOPSIS
    cat [OPTION]... [FILE]...

DESCRIPTION
    Concatenate FILE(s) to standard output.

    With no FILE, or when FILE is -, read standard input.

    -A, --show-all
        equivalent to -vET
```

```
analyst@db9351ee59fc:~$ whatis cat
cat (1) - concatenate files and print on the standard output
analyst@db9351ee59fc:~$ man cat
CAT(1) User Commands CAT(1)
analyst@db9351ee59fc:~$ apropos -a first part file
head (1) - output the first part of files
analyst@db9351ee59fc:~$ man useradd
USERADD(8) System Management Commands USERADD(8)

NAME
    useradd - create a new user or update default new user information

SYNOPSIS
    useradd [options] LOGIN

    useradd -D

    useradd -D [options]
```

```
analyst@db9351ee59fc:~$ whatis rm
rm (1) - remove files or directories
analyst@db9351ee59fc:~$ whatis rmdir
rmdir (1) - remove empty directories
rmdir (2) - delete a directory
analyst@db9351ee59fc:~$ apropos -a create new group
groupadd (8) - create a new group
analyst@db9351ee59fc:~$
```

Conclusion

In this activity, I learned how to use Linux's built-in help resources.

I practiced `whatis` for quick command descriptions, `man` for detailed documentation, and `apropos` for finding commands through keywords.

These commands will be useful as I continue learning Linux and cybersecurity because they allow me to find information when I do not remember a command or one of its options.

Course Information

Google Cybersecurity Certificate

Course: Tools of the Trade: Linux and SQL

Activity: Get Help in the Command Line

Disclaimer

This document is part of my personal cybersecurity learning portfolio. The activity was completed for educational purposes as part of the Google Cybersecurity Certificate.